

$\therefore$ By Definition, $\begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -7 \\ -6 \\ 0 \\ 1 \end{bmatrix}$ is a basis for $\text{Nul } A$. $\square$

$$25. A = \begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 8 & 0 & 5 \\ 0 & 2 & 5 & 0 & -1 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution: It's easy to see that Row 1 Column 1, Row 2 Column 2 and Row 3 Column 4 are pivot positions. $\therefore$ Column 1, 2 and 4 are pivot columns. By Theorem 13, the columns $\begin{bmatrix} 1 \\ -1 \\ -2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 2 \\ 2 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 3 \\ 5 \\ -5 \end{bmatrix}$ are a basis for $\text{Col } A$.

$$\begin{bmatrix} 1 & 4 & 8 & 0 & 5 & 0 \\ 0 & 2 & 5 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{-2} \times \text{Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & -2 & 0 & 7 & 0 \\ 0 & 2 & 5 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row 2} = \text{Row 2}}$$

$$\begin{bmatrix} 1 & 0 & -2 & 0 & 7 & 0 \\ 0 & 1 & 5/2 & 0 & -1/2 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding system of equations are:

$$\begin{aligned} x_1 - 2x_3 + 7x_5 &= 0 \\ x_2 + (5/2)x_3 - (1/2)x_5 &= 0 \\ x_4 + 4x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -5/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -7 \\ 1/2 \\ 0 \\ -4 \\ 1 \end{bmatrix}$$

Since x_3 and x_5 are free variables, $\mathbf{x}$ is a set of all possible linear combinations of the vectors $\begin{bmatrix} 2 \\ -5/2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -7 \\ 1/2 \\ 0 \\ -4 \\ 1 \end{bmatrix}$. Also they are linearly independent.

$\therefore$ By Definition, $\begin{bmatrix} 2 \\ -5/2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -7 \\ 1/2 \\ 0 \\ -4 \end{bmatrix}$ is a basis for $\text{Nul } A$. $\square$